Agent Communications Network Scenarios and Gap Analysis

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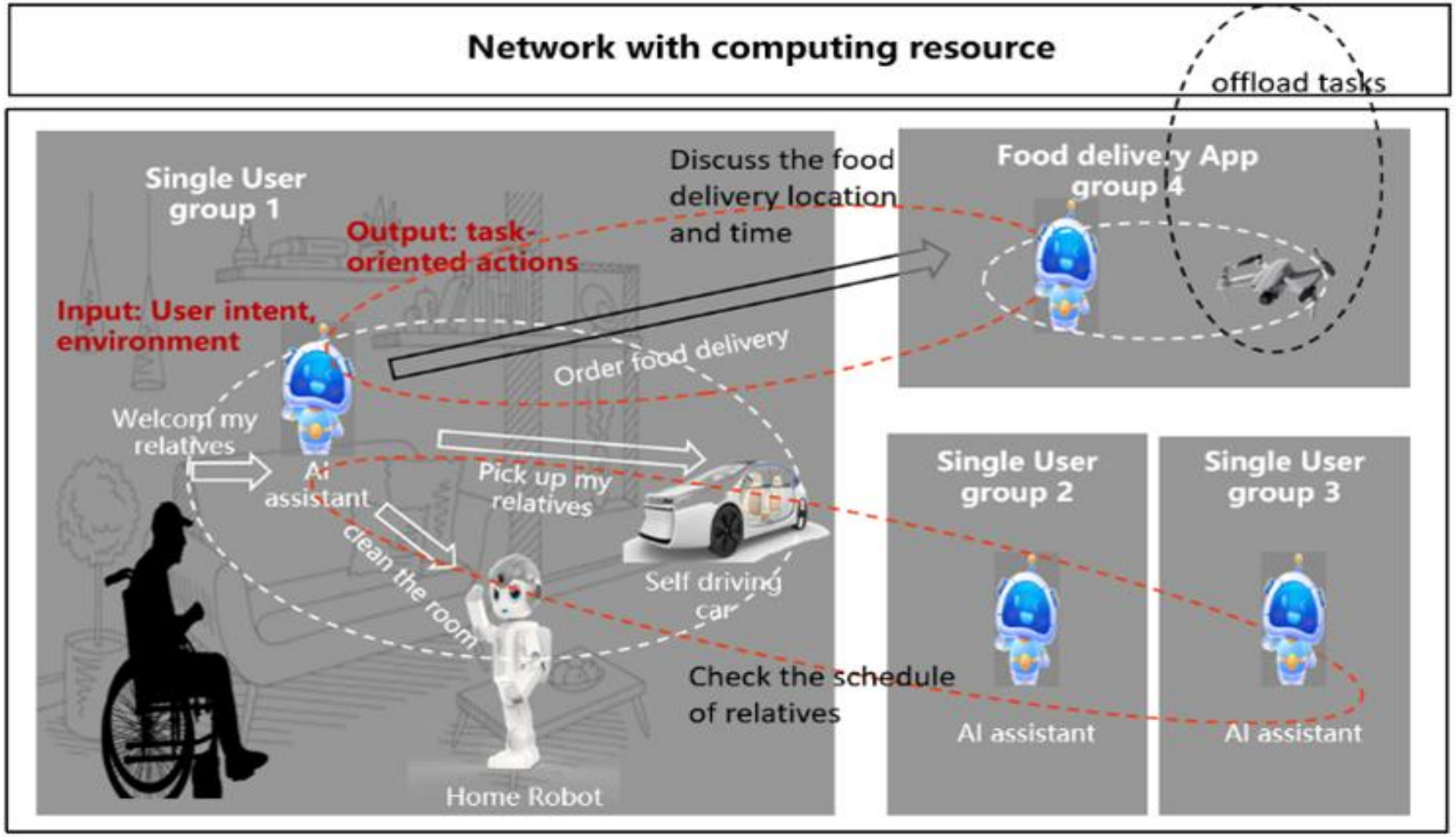
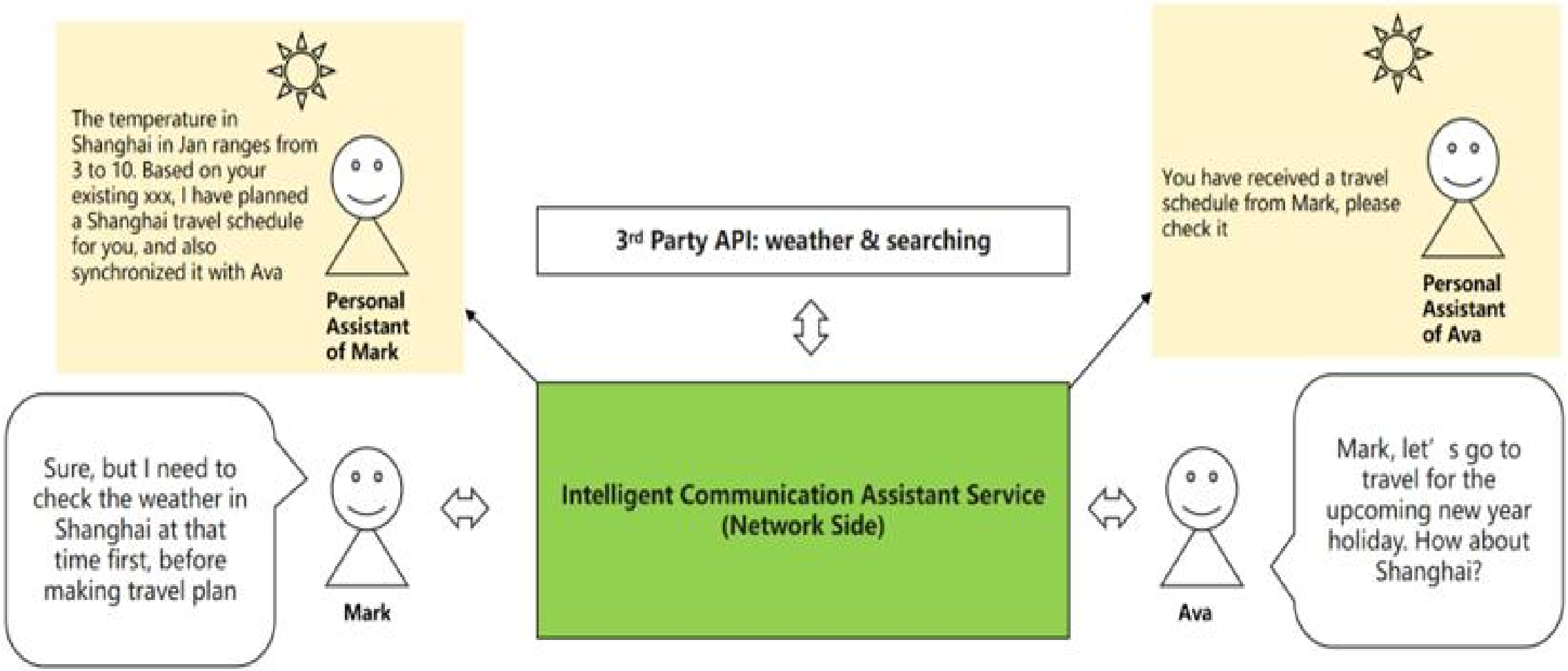
IETF 125

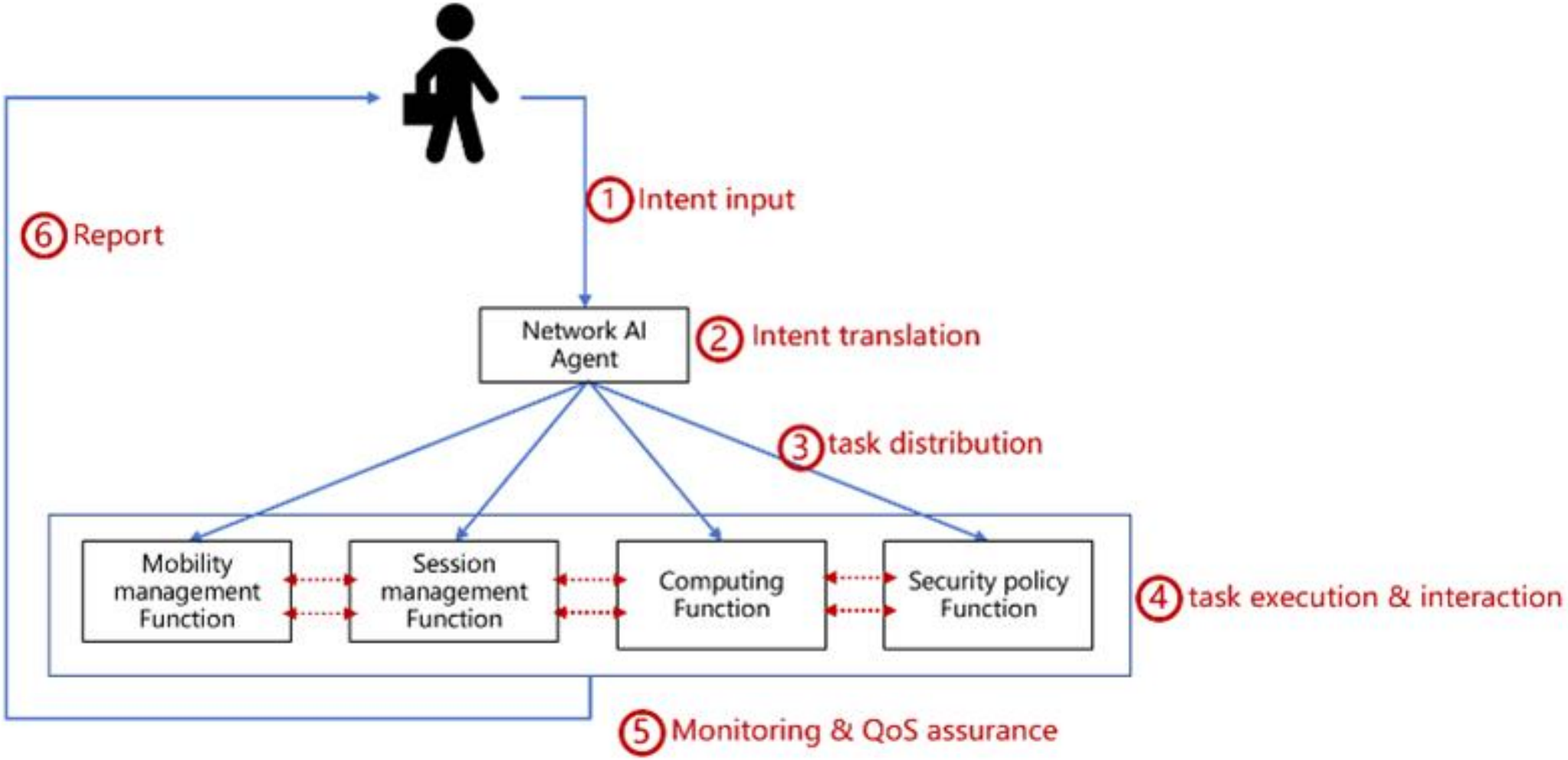
Potential Scenarios-End-side AI Agent(1/3)



Potential Scenarios-IMS AI Agent(2/3)



Potential Scenarios-Agentic Network Function(3/3)



Gap Analysis(1/2)

- **AI Agent Identity, Authentication, and Authorization**
  - Identity lifecycle management(registration, issuance, etc)
  - User/device binding
  - Delegated authorization
- **AI Agent Discovery**
  - Naming
  - Dynamic registration and information update
  - Mapping
  - Skill representation

Gap Analysis(2/2)

- **Session Management**
  - Session state semantics
  - Session state resumption
  - Native session timeout
- **Multi-modal Transport**
  - Latency bounded transmission
  - Message Ordering
  - Loss handling
  - Priority scheduling

Observations During IETF125

- Observations come from CATALIST BoF, WG meetings, planery, side meetings, hackathons, mailing list discussions, and F2F discussions.
- Willingness: enough people interested in AI agent comminucations, and think IETF should take the role for protocol work.
- Directions and work streams:
  - Security (Identity, AuthN, AuthZ, Auditing)
  - Discovery (Naming, mapping, selection)
  - Communications (Agent to agent/tool/user)
- Background knowledge unmatching -> need more coordination
- Use cases taxonomy, terminology, protocol gap analysis, framework design are prior before solutions (even though there are already solution candidates)